

# **L2b:** Yield, Duration, Convexity, and the Yield Curve

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CHEME 5660 Cornell University

# Disclaimer and Risks

**This material is for training and information only.** It does not provide investment advice or recommend any security, derivative, trade, or strategy.

**Trading involves risk.** Past results do not guarantee future results. Any risk you take should fit your resources, experience, and tolerance for loss.

**Your decisions are your responsibility.** Base them on your own objectives, financial position, risk tolerance, and liquidity needs.

# Scenario

Suppose you manage a \$500 million Treasury portfolio. The Federal Reserve meets today, and the market expects a possible 50-basis-point move in the policy rate.

If yields move, the value of the portfolio will move with them, and your clients will want to know **by how much**.

The estimate depends on the timing of the portfolio's cash flows and on the change in each security's yield. We will use the **slope** and **curvature** of the price-yield curve to compute the price response.

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**Policy rate and yield:** A half-point move in the *policy rate* does not imply a half-point move in *your security's yield*.

# Learning Objectives

We start with the L2a pricing model and study how fixed-income prices respond when yields change.

## Today's concepts

- **Price and yield:** Explain why a higher yield lowers the value of fixed positive cash flows and why the price-yield curve is convex.
- **Duration and convexity:** Use both measures to estimate a price change for a small change in yield, and state the units of each measure.
- **Yield measures:** Distinguish yield to maturity, par yield, spot rates, and maturity-specific discount factors.

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Companion notebook: [L2b lecture notebook](#)

# Lecture and Example Notebooks

**Lecture:** CHEME-5660-L2b-Lecture-Yield-Duration-Convexity-Curve-Fall-2026.ipynb

**Example:** CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb

The example prices a coupon Treasury security and computes the price response to changes in yield and coupon rate.

The advanced/ directory contains optional notebooks on Malkiel's theorems and the yield curve.

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**Link:** <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

Treasury prices change when market yields change.

# Are Treasury Securities Risk-Free?

**Not exactly.** U.S. Treasury securities have low default risk, but they are still exposed to interest-rate risk.

- **Default risk:** The issuer fails to make a promised payment. This risk is low for U.S. Treasury securities, although sovereign defaults do occur.
- **Interest-rate risk:** A change in market yields changes the value of fixed promised cash flows. The market value can fall even if the security is not sold.

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**Focus:** We focus on interest-rate risk because we can compute how a change in yield changes the price.

# Silicon Valley Bank and Interest-Rate Risk

Silicon Valley Bank held many **long-maturity, fixed-rate** government securities. Their market values fell when interest rates rose in 2022 and 2023.

- Many of the losses remained unrealized while the bank held the securities.
- In March 2023, uninsured depositors withdrew funds. To meet those withdrawals, SVB sold securities, realized losses, and announced a capital raise.
- The FDIC took control on March 10, 2023, when the bank had roughly \$209 billion in assets. At the time, it was the second-largest failure of an insured bank on record; it is now the third-largest.

The decline in asset values and the deposit withdrawals reinforced each other. Interest-rate risk was important, but it was not the only cause of the failure.

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**Overview:** [the collapse of Silicon Valley Bank, from PBS](#)



# Malkiel's Bond-Pricing Theorems

Malkiel's five theorems describe the prices of **option-free** bonds with fixed positive cash flows. Each comparison holds the other contract terms fixed. The maturity comparisons use bonds priced **at par**, so  $c = y$ .

1. The price falls when the yield rises.
2. A longer-maturity bond is more sensitive to a change in yield.
3. Each additional period adds less sensitivity than the previous one.
4. A decrease in yield raises the price more than an equal increase in yield lowers it.
5. A higher coupon rate reduces the percentage price response to a given change in yield.

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**Reference:** Malkiel (1962), *Expectations, Bond Prices, and the Term Structure of Interest Rates*, QJE 76(2), 197–218.

# Treasury Bill Pricing

A Treasury bill has one cash flow: the face value  $V_P$  paid at maturity.

Let  $T$  be the remaining time to maturity in years,  $n$  the number of compounding periods per year, and  $y$  the nominal annual yield. From L2a, the price is:

$$V_B = \underbrace{(1 + y/n)^{-nT}}_{\mathcal{D}_{T,0}^{-1}(y;n)} V_P.$$

A larger yield or a longer maturity discounts the face value more heavily. We assume  $V_P > 0$ ,  $T > 0$ , and  $y > 0$  unless stated otherwise.

# Price Sensitivity of a Treasury Bill

A bill has no coupon, so  $c = 0$  and  $dc = 0$ . Holding  $V_P$  and  $n$  fixed, its price changes with the yield  $y$  and the remaining maturity  $T$ :

$$dV_B = - \underbrace{V_P (1 + y/n)^{-nT} \left( \frac{T}{1 + y/n} \right)}_{V_{B,y}} dy - \underbrace{V_P (1 + y/n)^{-nT} (n \log(1 + y/n))}_{V_{B,T}} dT.$$

The quantities  $V_{B,y}$  and  $V_{B,T}$  are positive sensitivity magnitudes. The minus signs show that the price falls when either the yield or the remaining maturity increases, provided  $y > 0$ .

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**No  $dc$  term:** The contract fixes  $dc = 0$ . The derivative  $\partial V_B / \partial c$  is not zero, even at  $c = 0$ .

# Interpreting a Change in Maturity

A change in  $T$  can describe either a comparison between contracts or the passage of time for one contract:

- **Compare two contracts:** A 6-month bill and a 12-month bill have different contractual maturities. For a coupon bond, a longer maturity can also add coupon payments.
- **Hold one contract over time:** As calendar time advances, the remaining maturity falls, so  $dT = -dt$ . For a coupon bond, this change is smooth only between coupon dates.

# Treasury Bill Sensitivities

The sensitivity magnitudes for yield and maturity are:

$$V_{B,y} = \underbrace{V_P(1 + y/n)^{-nT}}_{=V_B} \underbrace{\left(\frac{T}{1 + y/n}\right)}_{>0}, \quad V_{B,T} = \underbrace{V_P(1 + y/n)^{-nT}}_{=V_B} \underbrace{(n \log(1 + y/n))}_{=g_y},$$

where  $g_y := n \log(1 + y/n)$  is the continuously compounded rate equivalent to  $y$ .

- $V_{B,y} > 0$  for  $T > 0$  and  $1 + y/n > 0$ .
- $V_{B,T} > 0$  requires  $y > 0$ .
- At  $y = 0$ , maturity does not affect price. For  $-n < y < 0$ , a longer bill has a higher price.

# How Yield and Time Affect a Treasury Bill

## Effect of a higher yield

A higher yield discounts the face value more heavily, so  $y \uparrow \Rightarrow V_B \downarrow$ . If a 52-week bill's yield rises from 2% to 3%, its market price falls.

## Effect of time passing

At a fixed positive yield, the price rises as the bill approaches maturity because fewer discounting periods remain:  $T \downarrow \Rightarrow V_B \uparrow$ . At  $y = 0$ , the discount factor is one, so the price does not depend on maturity.

# Optional Treasury Bill Example

The required lecture continues with coupon-bearing notes and bonds. The optional bill notebook examines maturity sensitivity in more detail.

**Question:** How does the price sensitivity of a zero-coupon bill change with its remaining maturity?

**Notebook:** The Treasury bill maturity example in advanced/malkiel/.

# Coupon-Bearing Treasury Notes and Bonds

A coupon-bearing Treasury security pays coupons before maturity and returns the face value at maturity. Let  $c$  be the annual coupon rate,  $C := (c/n)V_P$  the coupon paid each period, and  $N := nT$  the number of remaining payment dates. The L2a pricing model is:

$$V_B = \mathcal{D}_{N,0}^{-1}(y; n) V_P + C \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}(y; n), \quad \mathcal{D}_{j,0}^{-1}(y; n) = (1 + y/n)^{-j}.$$

The price has two parts:

- the discounted principal payment  $V_P$  at date  $N$ ;
- the discounted coupon payments  $C$  at dates  $j = 1, \dots, N$ .



# Yield and Coupon Sensitivity

Hold the payment schedule fixed and differentiate:

$$\frac{\partial V_B}{\partial y} = -\frac{1}{n(1 + y/n)} \left[ NV_P \mathcal{D}_{N,0}^{-1}(y; n) + C \sum_{j=1}^N j \mathcal{D}_{j,0}^{-1}(y; n) \right] < 0,$$

$$\frac{\partial V_B}{\partial c} = \frac{V_P}{n} \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}(y; n) > 0.$$

A higher yield discounts every payment more heavily, so the price falls. A higher coupon rate increases every coupon payment, so the price rises.

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**Schedule length:**  $N$  is a discrete index. We do not differentiate a finite sum with upper limit  $nT$  as if  $T$  were continuous.

# How Yield, Coupon, and Maturity Affect Price

- **Yield:**  $y \uparrow \Rightarrow V_B \downarrow$  because every coupon and the principal payment are discounted more heavily.
- **Coupon rate:**  $c \uparrow \Rightarrow V_B \uparrow$ , but not in direct proportion. Doubling  $c$  doubles the coupon stream, not the principal payment.
- **Remaining maturity:** The direction of the price change depends on  $c - y$ . As maturity shortens, a discount bond ( $c < y$ ) rises toward par, a premium bond ( $c > y$ ) falls toward par, and a par bond ( $c = y$ ) stays at par on coupon dates.

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**Maturity comparison:** Malkiel's result concerns *sensitivity to a yield change*, not the price level.

# A General Cash-Flow Pricing Model

For an **option-free** security, the promised cash flows do not change when yields change. Conventional noncallable Treasury securities satisfy this assumption.

Let  $X_j \geq 0$  be the cash flow at period  $j = 1, \dots, N$ , with at least one positive cash flow. The price is:

$$V_B(y) = \sum_{j=1}^N X_j \mathcal{D}_{j,0}^{-1}(y; n).$$

- **Note or bond:**  $X_j = C$  for  $j < N$  and  $X_N = C + V_P$ .
- **Bill:**  $X_j = 0$  for  $j < N$  and  $X_N = V_P$ .

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**Outside this model:** callable, inflation-indexed, and mortgage-backed securities can have cash flows that change with rates or states.

# The Price-Yield Relationship

Holding the cash flows  $X_j$  and the number of payments  $N$  fixed gives:

$$\frac{dV_B}{dy} = -\frac{1}{n} \sum_{j=1}^N j X_j (1 + y/n)^{-j-1} < 0,$$

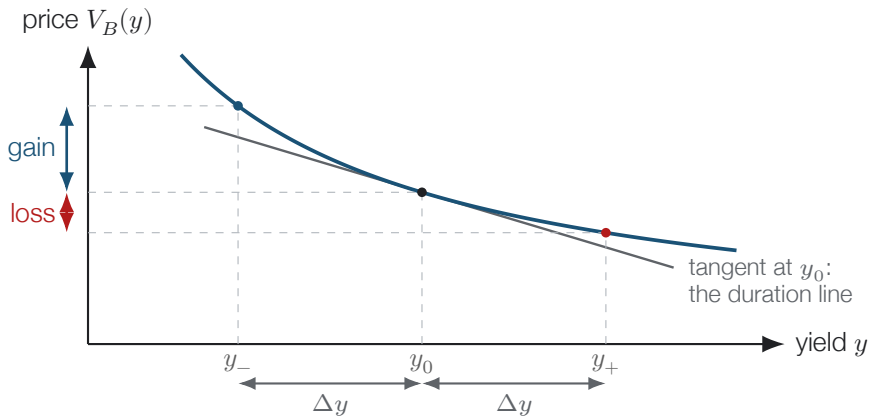
$$\frac{d^2V_B}{dy^2} = \frac{1}{n^2} \sum_{j=1}^N j(j+1) X_j (1 + y/n)^{-j-2} > 0.$$

- $dV_B/dy < 0$ : The price decreases as the yield increases.
- $d^2V_B/dy^2 > 0$ : The price-yield curve is strictly convex.

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**Sign conditions:**  $1 + y/n > 0$ , every  $X_j \geq 0$ , and at least one  $X_j > 0$ .

# Convexity and Asymmetric Price Changes



For equal changes in yield, the **gain** from a decrease is larger than the **loss** from an increase. The tangent line gives the duration estimate, while convexity accounts for the curvature.

# Modified Duration and Convexity

The price derivatives measure dollar sensitivity, so they depend on the size of the position. Dividing by the current price  $V_B(y) > 0$  gives percentage sensitivity measures that can be compared across securities:

$$D_{\text{mod}}(y) := -\frac{1}{V_B(y)} \frac{dV_B}{dy}, \quad K(y) := \frac{1}{V_B(y)} \frac{d^2V_B}{dy^2}.$$

- **Modified duration**  $D_{\text{mod}}$  measures the local percentage slope of price with respect to yield.
- **Convexity**  $K$  measures the local percentage curvature.
- If  $y$  is a decimal annual yield,  $D_{\text{mod}}$  has units of years and  $K$  has units of years squared.

# Macaulay and Modified Duration

Macaulay duration is the present-value-weighted average time to payment. Modified duration converts that measure of cash-flow timing into local price sensitivity:

$$D_{\text{mod}} = \frac{D_{\text{mac}}}{1 + y/n}.$$

A bill makes its only payment at maturity, so:

$$D_{\text{mac}} = T, \quad D_{\text{mod}} = \frac{T}{1 + y/n} = \frac{V_{B,y}}{V_B}.$$

# Estimating a Price Change

For a small yield change  $\Delta y$ , the second-order Taylor approximation is:

$$\frac{\Delta V_B}{V_B} \approx \underbrace{-D_{\text{mod}} \Delta y}_{\text{slope}} + \underbrace{\frac{1}{2}K(\Delta y)^2}_{\text{curvature}}.$$

Because  $K > 0$ , the convexity term is positive for either sign of  $\Delta y$ . It increases the estimated gain when the yield falls and reduces the estimated loss when the yield rises.

Use decimal yield changes: one basis point is  $10^{-4}$ , so 50 basis points is  $\Delta y = 0.0050$ .

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**Policy rate and yield:** A 50-basis-point move in the *policy rate* does not imply  $\Delta y = 0.0050$  for a given security.



## Example: A 50-Basis-Point Yield Increase

Suppose a note has  $D_{\text{mod}} = 7.2$  years and  $K = 65$  years<sup>2</sup>. If its yield rises by 50 basis points, then  $\Delta y = 0.0050$ :

- The duration term is  $-7.2 \times 0.0050 = -0.0360$ .
- The convexity term is  $\frac{1}{2} \times 65 \times (0.0050)^2 = +0.0008$ .
- The estimated price change is **-3.52%**, compared with -3.60% using duration alone.

Duration and convexity are **local** measures. For a large yield change, reprice the full cash-flow schedule at the new yield.

# Portfolio Duration

- Duration places securities with different coupons, maturities, and prices on the same sensitivity scale.
- Portfolio duration is the market-value-weighted average of the durations of the individual positions.
- Offsetting positions can reduce net duration. Funding long-duration assets with short-duration liabilities creates a duration mismatch.

These calculations assume that each position experiences the stated yield change. If different maturities move by different amounts, we need maturity-specific yield changes.

# Duration of a Par Bond

To compare sensitivity across maturities, consider option-free bonds with fixed positive cash flows and hold the other terms fixed. Each bond starts at **par**, so  $c = y$ .

Sum the coupon-bond price, differentiate at fixed  $c$ , and then set  $c = y$ :

$$\frac{V_B}{V_P} = \frac{c}{y} + \left(1 - \frac{c}{y}\right) (1 + y/n)^{-N} \quad \Rightarrow \quad D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}.$$

The order of these steps matters. If we set  $c = y$  before differentiating, the price remains at  $V_P$  along the par curve and the derivative is zero. We assume  $y > 0$ .

# Malkiel's Theorems 1 and 2

## 1. The price falls when the yield rises

The result  $dV_B/dy < 0$ , or equivalently  $D_{\text{mod}} > 0$ , follows because a higher yield discounts every positive payment more heavily.

## 2. A longer par bond is more sensitive to yield

$$D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}$$

increases with  $N$ . Because each bond starts at  $V_B = V_P$ , the dollar sensitivity  $-dV_B/dy = V_P D_{\text{mod}}(N)$  also increases with maturity.

# Malkiel's Theorem 3

Adding one payment period changes modified duration by:

$$D_{\text{mod}}(N + 1) - D_{\text{mod}}(N) = \frac{(1 + y/n)^{-N-1}}{n} > 0.$$

The next increment equals the previous increment multiplied by:

$$(1 + y/n)^{-1} < 1.$$

Modified duration therefore increases with maturity, but each additional period adds less than the previous one. As  $N \rightarrow \infty$ , modified duration approaches  $1/y$ .

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**Limit:**  $1/y$  is the modified duration of a perpetuity.

# Malkiel's Theorem 4

Start at the same yield and compare equal changes  $\pm\Delta y$ .

$$\frac{\Delta V_B}{V_B} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}K(\Delta y)^2$$

The duration term changes sign when the direction of the yield change changes. The positive convexity term does not. Therefore, a yield decrease raises the price more than an equal yield increase lowers it.

The direction of the curvature matters. A concave price-yield curve would produce the opposite asymmetry.

# Malkiel's Theorem 5

A higher coupon shifts more of the bond's present value to earlier payment dates. This lowers Macaulay duration and therefore lowers  $D_{\text{mod}}$ .

This comparison requires  $N \geq 2$  and concerns **percentage sensitivity**. Dollar sensitivity can still increase because a higher coupon raises the bond's price:

$$-\frac{dV_B}{dy} = V_B D_{\text{mod}}.$$

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**Edge case:** at  $N = 1$ , every payment arrives on one date and  $D_{\text{mod}} = 1/[n(1 + y/n)]$ , regardless of coupon size.

# Why Theorems 2 and 3 Compare Bonds at Par

Away from par, percentage sensitivity and dollar sensitivity can rank bonds differently. For a zero-coupon bill,  $D_{\text{mod}} = T/(1 + y/n)$  continues to grow with maturity, but dollar sensitivity contains one factor that grows and another that shrinks:

$$-\frac{dV_B}{dy} = V_P \underbrace{\frac{T}{1 + y/n}}_{\text{growing}} \underbrace{(1 + y/n)^{-nT}}_{\text{shrinking}}.$$

With  $V_P = 100$ ,  $n = 1$ , and  $y = 5\%$ , dollar sensitivity increases at first, reaches a maximum near  $T \approx 20.5$  years, and then decreases. It is about 718 USD per unit of yield at the maximum and about 541 at  $T = 40$ .

The 40-year bill is more sensitive in **percentage** terms but less sensitive in **dollars**. Starting both bonds at par makes the rankings agree.

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**Example:** CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb, which tests Theorems 4 and 5.



# Yield Measures and the Yield Curve

- **Yield to maturity** is the single rate that reproduces a security's market price from its promised cash flows.
- A **spot rate** applies to one payment date, so the spot curve provides a separate discount factor for each date.
- A **par yield** is the coupon rate that prices a hypothetical bond at face value under the spot curve. The published Treasury yield curve is a par yield curve.

Let  $D(0, t_j)$  be the time-0 discount factor for a payment at date  $t_j$ . Then:

$$V_B = \sum_{j=1}^N X_j D(0, t_j).$$

The level, slope, and curvature of the yield curve describe how discounting varies with maturity. They do not, by themselves, determine future short rates.

# Inverted Yield Curves

A yield curve is **inverted** when a long-maturity yield is below a short-maturity yield.

An inverted spread such as **10-year minus 3-month** has preceded every U.S. recession of the past half century.

This spread does not tell us when a recession will begin. False positives occur, lead times range from months to years, and the result depends on the maturity pair. The optional par-yield example computes the spread from Treasury data.

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**Notebook:** [advanced/yield\\_curve/CHEME-5660-L3a-ParYieldCurves-Example-Fall-2026.ipynb](#)

# Optional Advanced Material

**Malkiel:** [advanced/malkiel/](#)

The Malkiel notebooks study Treasury-bill maturity data and derive a closed-form proof of Theorems 2 and 3.

**Yield curve:** [advanced/yield\\_curve/](#)

The yield-curve notebooks compute investment yield, solve for yield to maturity with the secant method, and construct par yield curves.

**STRIPS:** [advanced/strips/](#)

The STRIPS notebooks infer spot, par, and implied forward rates from Treasury STRIPS prices.

These notebooks are optional and are not prerequisites for L3a.

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**Notebooks:** [lectures/week-2/L2b/advanced](#), also linked from the lecture notebook.

# Summary

We used cash-flow pricing to measure the interest-rate risk of fixed-income securities.

## Key takeaways

- **Price and yield move in opposite directions.** For fixed positive cash flows, the price-yield curve is decreasing and convex.
- **Duration and convexity estimate small price changes.** The sensitivity depends on the maturity, coupon, and current yield.
- **Different yield measures answer different questions.** Yield to maturity summarizes one security, spot rates price individual payment dates, and par yields summarize the spot curve as coupon rates.

How can **duration and convexity** help us manage the interest-rate risk of a bond portfolio?